

PROJECT 3

SQL ANALYSIS REPORT

This project focused on analyzing the Online Retail Dataset using Microsoft SQL Server and SQL Server Management Studio. The objective of the project was to apply SQL querying, data cleaning, filtering, aggregation, and reporting techniques to extract meaningful business insights from retail transaction data.

DATABASE SETUP AND DATA CLEANING

The dataset was successfully imported into SQL Server from a CSV file. Data cleaning and transformation processes were carried out to improve data quality and prepare the dataset for analysis. Blank values in the Coupon Code column were identified and replaced with “No Coupon” to maintain consistency. Additional columns for Order_Year and Order_Month were created and populated using SQL date functions to support time-based analysis. Data types for all columns were also verified and adjusted where necessary.

SQL SKILLS AND TECHNIQUES APPLIED

The project demonstrated practical use of core SQL concepts and analytical functions, including:

- SELECT statements for data extraction
- WHERE clause for filtering records
- ORDER BY for sorting results
- GROUP BY for aggregation
- HAVING clause for filtering grouped data

- Aggregate Functions:
 - SUM()
 - AVG()
 - COUNT()

Additional SQL operations included data updates, date extraction functions, and conditional filtering for business analysis.

KEY FINDINGS AND BUSINESS INSIGHTS

OVERALL METRICS PERFORMANCE

- Total Orders: 1,200
- Total Revenue: 1,264,761.96
- Average Order Value: 1,053.97
- Average Quantity Ordered: 2 items
- Cancelled Orders: 251

CUSTOMER AND PAYMENT INSIGHTS

- Most Used Payment Method: Online Payments (259 transactions)
- Highest Revenue Payment Method: Credit Card (263,847.63)
- Top Referral Source: Instagram (260 referrals)
- Highest Revenue Referral Source: Instagram (275,285.45)

PRODUCT PERFORMANCE

The analysis identified strong-performing products across the dataset:

- Top 3 Products by Revenue:
 1. Chair — 195,620.11

2. Printer — 195,612.61

3. Laptop — 192,126.56

Products with sales greater than 2000 included Chair, Desk, Laptop, Monitor, Phone, Printer, and Tablet.

The product with the highest individual total price was:

- Tablet
- Coupon Code: SAVE10
- Total Price: 3,456.40

The product with the lowest recorded price was:

- Phone
- Referral Source: Email
- Total Price: 11.39

REVENUE ANALYSIS

REVENUE BY PAYMENT METHOD

Payment Method	Total Revenue
Credit Card	263,847.63
Online	262,442.94
Cash	259,786.29
Gift Card	246,323.92
Debit Card	232,361.18

REVENUE BY REFERRAL SOURCE

Referral Source	Total Revenue
Instagram	275,285.45
Email	261,808.55
Google	250,441.48
Facebook	250,410.90
Referral	226,815.58

CONCLUSION

This project demonstrated practical SQL and data analysis skills through the use of a real-world retail dataset. SQL queries were effectively used to clean, organize, analyze, and summarize transactional data into actionable business insights. The analysis provided visibility into customer behavior, product performance, payment trends, and revenue distribution. Overall, the project strengthened foundational SQL skills and showcased the ability to transform raw data into meaningful business intelligence suitable for decision-making and reporting purposes.

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